Higher National Diploma in Engineering (Electrical)
Fourth Year, First Semester Examination – 2022
EE 3204– Electrical Energy Management

Instructions for Candidates:
Answer **Five (05)** questions.
Each question carries equal marks

No. of questions:06
No. of pages : 04
Time : **03 Hours**

Q1.

(a) Write any **Four (04)** differences between shell type & Core type Transformer?

(04 Marks)

(b) Define the following losses of tranformer;

i.) Copper Loss
ii.) Eddy current loss
iii.) Iron Loss

(06 Marks)

(c) Single phase 50Hz transformer has 100 turns on the primary and 500 on the secondary winding. Net cross section area of the core is 200 cm^2. If the primary supply is 240 V AC 50Hz supply

i.) Find the induced e.m.f in the secondary
ii.) Find the maximum value of flux density in the core

(06 Marks)

(d) 400/200 V transformer takes 1.5A at 0.4 lagging p.F on no load. If the secondary supplies 50A current at 0.8 p.f lagging, determine the primary current and input power factor?

(04 Marks)

[Total 20 Marks]

Q2.

(a) Briefly explain the following terms and importance of them for lighting design

i.) Color rendering & color rendering index (CRI)

ii.) Contrast (04 Marks)

(b) List out the type of factors affecting on light designing system? (04 marks)

(c) What are the types of lighting schemes? Briefly describe them (05 Marks)

(d) What are the types of lamps use for lighting? Explain the construction & working principle of **incandescent lamp** and also discuss ant two advantage & disadvantage of that lamp? (07 Marks)

[Total 20 Marks]

Q3.

(a) Define the following terms,

i.) Luminous flux

ii.) Luminous Intensity

iii.) Illuminance

(06 Marks)

(b) State the **four (04)** Zone of inside the Tunnel lighting scheme, with diagram? (04 marks)

(c) A light is placed 2 m above the ground and its candle power is 120 cos θ in any downward direction making an angle θ with the vertical. If P and Q are two points on the ground, P being vertically directly under the light and the distance PQ being 2.6 m, calculate.

i.) Illumination of the ground at P

ii.) Illumination at Q. (04 Marks)

(d) Define **luminous efficacy** and briefly describe the importance of that parameter? (02 marks)

(e) A 250 V lamp has a total flux of 3000 lumens and takes a current of 0.8 A from 250 V mains. Find luminous efficacy of the lamp?

(04 Marks)

[Total 20 Marks]

Q4.

(a) Briefly explain the term **tariff**? State any Five (05) characteristics of a Tariff?

(06 Marks)

(b) Discuss the various methods for power factor improvement?

(04 marks)

(c) Write short notes on the following:

i.) Two-part tariff.
ii.) Power factor tariff.
iii.) Three-part tariff.

(05 marks)

(d) A consumer has a maximum demand of 200 kW at 40% load factor. If the tariff is Rs. 100 per kW of maximum demand plus 10 cents per kWh, find the overall cost per kWh.

(05 Marks)

[Total 20 Marks]

Q5.

(a) State the difference between slip ring rotor and cage rotor of an induction motor?

(03Marks)

(b) List out any **four (04)** starting method of Induction Motor?

(04 Marks)

(c)

i.) Write an expression for the slip of an induction motor.

(02 marks)

ii.) List out **three** the methods of speed control of cage type 3-phase induction motor?

(03 Marks)

(d) A 450-V, 50-Hz, 37kW, three-phase induction motor is drawing 60 A at 0.85 PF lagging. The stator copper losses are 2 kW, and the rotor copper losses are 700 W. The friction and windage losses are 600 W, the core losses are 1500 W, and the stray losses are negligible.

Find the following quantities:

i. Airgap power
ii. Converter power to the mechanical energy
iii. Output power
iv. Efficiency of the motor

(08 Marks)

[Total 20 Marks]

Q6.

(a) What are the needs of energy planning and audit?

(03 Marks)

(b) What are the types of Energy audit and explain walkthrough energy audit?

(04 Marks)

(c) Define **Demand side Management** and describe in details about the main objectives of DSM and what are the activities used for DSM. Give examples for such activities

(06 Marks)

(d) Explain the role of energy auditor? Describe the procedure for energy audit, step by step?

(07 Marks)

[Total 20 Marks]